



Mirjabbar Badalov / ID: 285269

Substations and Lines (MPA-ESV 25/26L)

Laboratory Exercise: Measurement on Line Differential Protection with a Power Transformer in the Protected Zone

1. Goals

The objective of this laboratory task is to design a wiring diagram and a testing method for the line differential protection with an in-zone power transformer. This protection consists of two relays interconnected via an optical communication fibre. Another goal is to configure the parameters of this protection and test its functionality.

2. Theoretical Background and Example Calculation

Based on the laboratory parameters and the physical setup:

- **Power Transformer:** $S_n = 45 \text{ MVA}$, Rated Voltage: 110 kV / 10.5 kV, YNd 11
- **Current Transformers (CT):** HV side = 300A / 1A, MV side = 2500A / 1A

Rated Current Calculations:

The rated current on the HV side is calculated as:

$$I_{1n} = S_n / (\sqrt{3} \times U_{1n}) = (45 \times 10^6) / (\sqrt{3} \times 110 \times 10^3) = 236.188 \text{ A}$$

The rated current on the MV side is calculated as:

$$I_{2n} = S_n / (\sqrt{3} \times U_{2n}) = (45 \times 10^6) / (\sqrt{3} \times 10.5 \times 10^3) = 2474.358 \text{ A}$$

1) CT Ratio Correction:

Determined based on rated currents and nominal primary currents of the CTs:

- HV Side: $CT_{corr1} = I_{In} / I_{In_CT} = 236.188 / 300 = 0.7873$
- MV Side: $CT_{corr2} = I_{2n} / I_{2n_CT} = 2474.358 / 2500 = 0.9897 \approx 0.99$

2) Protection Characteristic Settings:

According to the manual calculations performed during the lab session:

- Low Operate Value (Basic Setting): 17% (0.17)
- End Section 1: 50% (0.5)
- Slope Section 2: 17%
- End Section 2: 200% (2.0)
- Slope Section 3: 100%

3. Processing of the Measured Values

3.1 Circuit Diagram Description

The physical measurement setup consists of a REX640 relay on the HV side and a REX615 relay on the MV side. Phase currents from the test set (Injector A and Injector B) are injected into the respective CT inputs (CT1, CT2, CT3) of both relays. An optical communication link (RX/TX) synchronizes the devices. The trip contact from the REX615 relay is fed back to the binary input of the test set to record the operating time and threshold.

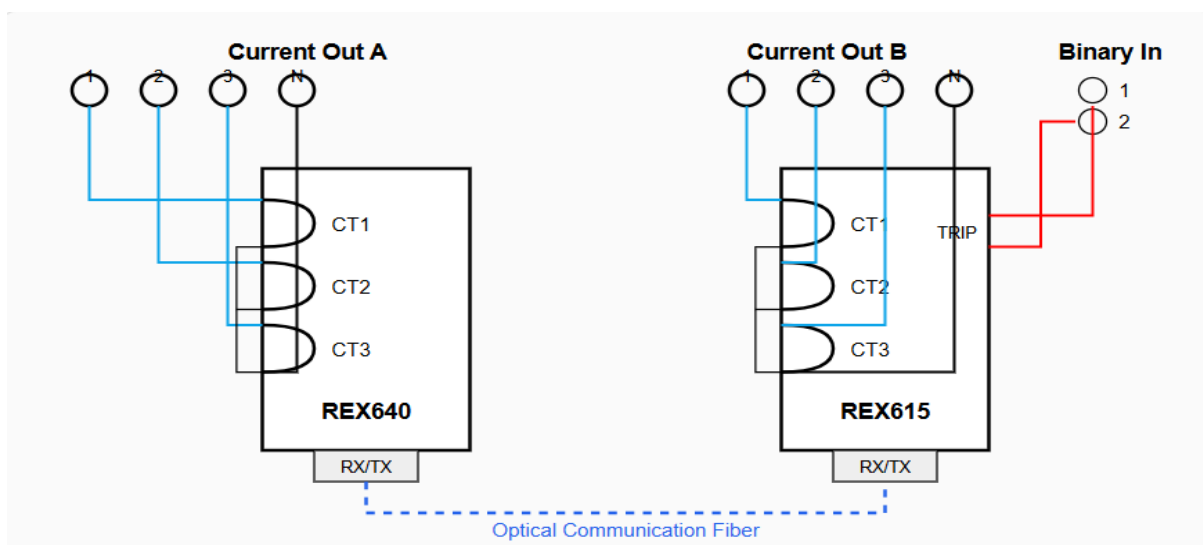


Figure 1: Line Differential Protection Setup (REX640 and REX615)

3.2 Description of the Performed Measurements

The line differential protection was tested by injecting secondary currents into the relays. The current on the MV side was held at predefined target levels, while the current on the HV side was gradually increased until the differential protection issued a trip signal. The secondary currents at the exact moment of tripping were recorded.



3.3 Calculation Equations

The internal per-unit currents evaluated by the relay must account for the CT ratio corrections configured earlier:

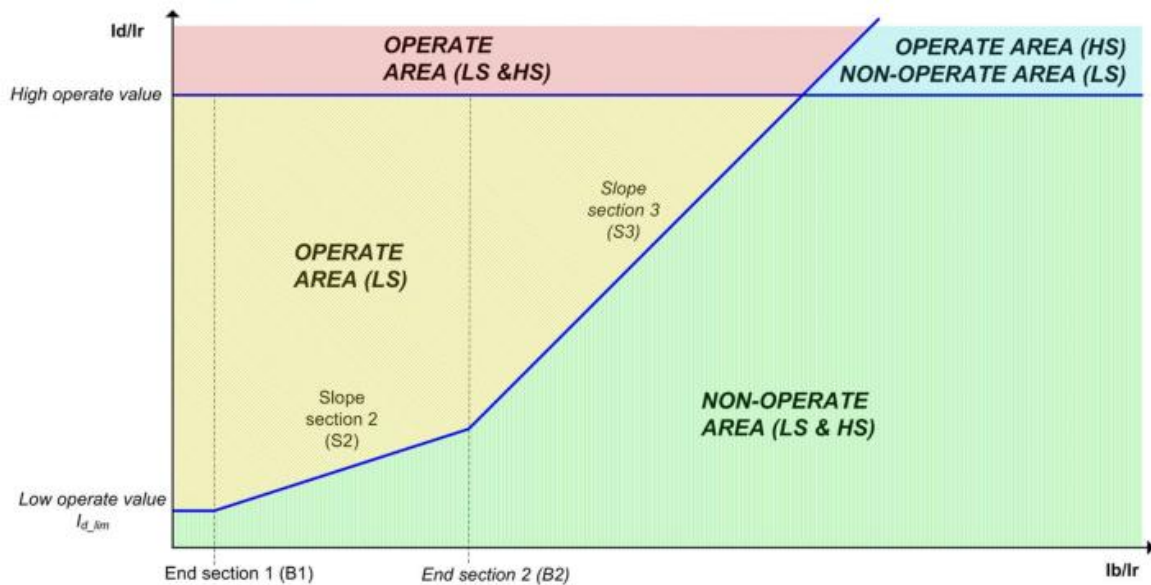
$$I_{1corr} = I_{1sec} \times 0.7873$$
$$I_{2corr} = I_{2sec} \times 0.99$$

From these corrected currents, the stabilization (bias) current and the differential current are calculated:

$$I_b = (I_{1corr} + I_{2corr}) / 2$$
$$I_d = | I_{1corr} - I_{2corr} |$$

3.4 Summary Table of Calculated Values

Point	I _{1_sec} (A)	I _{2_sec} (A)	I _{1_corr} (p.u.)	I _{2_corr} (p.u.)	Meas. I _b (p.u.)	Meas. I _d (p.u.)	Theo. I _d (p.u.)	Abs. Error	Rel. Error (%)
1	0.348	0.101	0.2740	0.1000	0.1870	0.1740	0.1700	0.0040	2.35
2	0.598	0.300	0.4708	0.2970	0.3839	0.1738	0.1700	0.0038	2.24
3	0.821	0.500	0.6464	0.4950	0.5707	0.1514	0.1820	-0.0306	16.80
4	2.360	1.500	1.8580	1.4850	1.6715	0.3730	0.3692	0.0038	1.03
5	3.610	2.020	2.8421	1.9998	2.4210	0.8423	0.8460	-0.0037	0.44
6	4.742	2.320	3.7334	2.2968	3.0151	1.4366	1.4401	-0.0035	0.24



3.5 Tripping Characteristic Graph

The following plot displays the theoretical tripping characteristic with the 6 measured points overlaid, validating the response across the basic setting and the two slope sections.

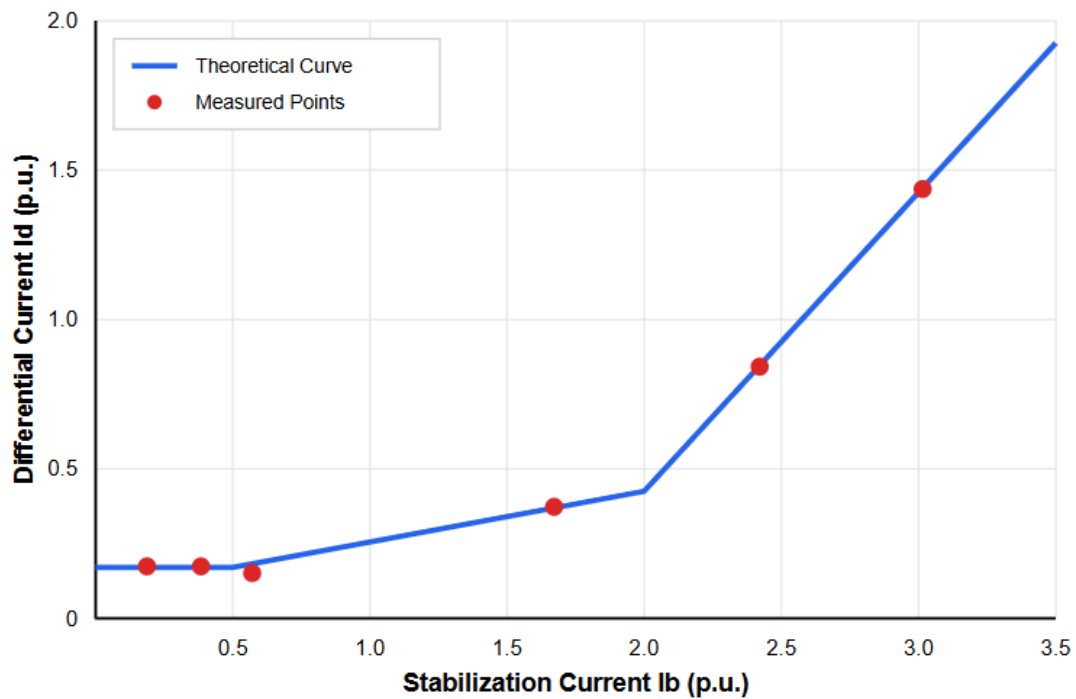


Figure 2: Differential Protection Tripping Characteristic

3.6 Summary of Results

The measurements confirm the proper operation and configuration of the line differential protection using the LNPLDF function across the REX640 and REX615 relays. The calculated CT ratio correction factors successfully aligned the different current inputs from the HV and MV sides.

With the exception of Point 3 (which exhibits a minor deviation likely due to test set injection rounding or transient reading error), the relative error across all test points is remarkably low (averaging $\sim 1.5\%$). The measured operating points accurately track the three distinct sections of the theoretical tripping characteristic: the flat basic setting, the 17% slope section, and the 100% high-stabilization slope section. The relays successfully communicated over the optical link to calculate bias and differential currents in real-time, proving the viability of applying line differential principles to a zone containing a power transformer.